MeetTask AI

AI Meeting Intelligence and Action Management Platform

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Project Overview

The **Secure AI Meeting Intelligence Platform** is an AI-powered application that assists organizations in managing meetings more efficiently. The platform records meetings, converts speech into text, and uses AI to generate Minutes of Meeting (MoM), summaries, key decisions, and action items. It also provides an integrated task management workflow that transforms approved action items into trackable tasks, which are assigned to employees through personalized dashboards, followed by email reminders and lifecycle tracking. The platform incorporates security mechanisms to ensure secure handling of confidential meeting information.

Problem Statement

AI-powered meeting assistants have improved the process of generating meeting transcripts, summaries, and action items. However, after extracting action items from meetings, organizations often rely on separate task management tools or manual processes to assign responsibilities, monitor progress, and follow up on completion. This separation between meeting documentation and task execution increases manual effort, creates workflow gaps, and reduces visibility into the status and history of tasks assigned during meetings.

Additionally, meetings often contain confidential organizational information that requires secure storage and controlled access. Meeting recordings, transcripts, and AI-generated outputs must be protected from unauthorized access, accidental disclosure, and misuse. AI-generated content may also require human verification before being distributed to ensure accuracy and reliability.

This project aims to bridge the gap between **AI-powered meeting intelligence and task execution** by providing a unified platform that generates meeting insights, manages the complete lifecycle of assigned tasks, and ensures secure handling of organizational meeting data.

Objectives

- Develop an AI-powered meeting assistant that automatically generates transcripts, Minutes of Meeting (MoM), summaries, decisions, and action items.
- Automatically assign extracted action items to employees through personalized dashboards.
- Send reminder notifications and track task completion until closure.
- Encrypt meeting recordings and AI-generated outputs before storage.
- Generate a unique AES-256 encryption key for every meeting.
- Implement Role-Based Access Control (RBAC) to restrict meeting access.
- Require OTP verification for meetings marked as **Confidential**.
- Maintain audit logs for every meeting access attempt.
- Allow managers to review and approve AI-generated outputs before they are published.

AI Capabilities

- **Speech-to-Text Transcription:** Converts recorded meeting audio into structured text using Whisper.
- **Meeting Summarization:** Uses NLP-based analysis to generate concise summaries and Minutes of Meeting (MoM).
- **Decision Extraction:** Identifies important decisions and conclusions discussed during meetings.

- **Action Item Extraction:** Analyzes meeting conversations to identify assigned tasks, responsible individuals, and deadlines.
- **Natural Language Understanding:** Processes meeting discussions to identify important information and transform unstructured conversations into structured outputs.

Proposed Architecture

1. **Meeting Creation:** The organizer creates a meeting, adds participants, and selects the confidentiality level of the meeting.
2. **Meeting Processing:** The meeting audio is recorded and converted into text using Speech-to-Text technology.
3. **AI Analysis:** AI analyzes the transcript and generates Minutes of Meeting (MoM), summaries, key decisions, and action items including assigned individuals and deadlines.
4. **Secure Storage & Review Workflow:** The generated content is securely stored in a **Pending Review** state. The meeting host/manager receives an email notification to review the AI-generated outputs.
5. **Manager Approval:** The manager reviews, edits, and approves the generated MoM and action items. Low-confidence or incorrect AI-generated information can be corrected before publication.
6. **Task Assignment & Tracking:** After approval, action items are converted into tasks and published to employee dashboards. Employees can update task status, while managers can monitor progress, receive completion updates, and view task history. Reminder notifications are automatically sent for pending tasks.
7. **Security & Access Management:** Meeting data is stored securely using encryption. Users access meetings according to their assigned roles through RBAC. Confidential meetings require OTP verification before access, and all access activities are recorded through audit logs.

Security & Governance Features

Security Concerns	Our Solution
AI-generated outputs may contain inaccuracies or hallucinated action items.	Managers review and approve AI-generated content before it is published.
Sensitive meeting data requires strong protection.	Every meeting is encrypted using AES-256 before being stored.
Confidential meetings require stronger authentication.	OTP verification is required before accessing confidential meetings.
Unauthorized users may access meeting data.	Role-Based Access Control (RBAC) restricts access based on user roles.
Organizations need accountability for sensitive information.	Every meeting access attempt is recorded in audit logs.
Sensitive information may appear in transcripts.	AI-assisted PII redaction masks sensitive information before transcripts are stored or shared.

Technologies Planned

Component	Technology
Speech-to-Text	Whisper (open-source)
AI Processing	Ollama (Llama/Gemma) or Groq API
Backend	Node.js (Express) or Django
Database	PostgreSQL or MongoDB
Encryption	Node crypto module / Python cryptography library
Frontend	React + Tailwind CSS
OPT/Email	Gmail SMTP or SendGrid (free tier)
Authentication	JWT

Expected Outcome

The project will deliver a working prototype that combines **Artificial Intelligence, Software Engineering, and Cybersecurity** to automate meeting understanding, convert discussions into trackable tasks, and securely manage confidential meeting information through encryption, role-based access, OTP verification, audit logging, and AI-assisted data protection.